



```
1  module ic(  
2      input wire [63:0] data,  
3      input wire [1:0] amount,  
4      input wire en,load,clk,  
5  
6      output reg [63:0] q  
7  );  
8  
9  always @(posedge clk)  
10 begin  
11     if (~en)  
12     begin  
13         case (amount)  
14             2'b00: q <= q << 8;  
15             2'b01: q <= q << 2;  
16             2'b10: q <= $signed(q) >>> 2;  
17             2'b11: q <= $signed(q) >>> 8;  
18         endcase  
19     end  
20     else if (load)  
21         q <= data;  
22 end  
23  
24 endmodule  
25
```

